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Review article

EcoHealth and One Health: A theory-focused review in response to calls for convergence

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ABSTRACT

Background: EcoHealth and One Health are two major approaches broadly aimed at understanding the links between human, animal, and environment health. There have been increasing calls for convergence between the two. If convergence is desired, greater clarity regarding the underlying theoretical assumptions of both approaches is required. This would also support integrated research to effectively address complex health issues at the human, animal and environment interface. To better understand the areas of overlap and alignment, we systematically compared and contrasted the theoretical assumptions of both approaches.

Objectives: We aimed to gain a more in-depth understanding of the ontological, epistemological and methodological underpinnings of EcoHealth and One Health in order to identify areas of difference and overlap, and consider the extent to which closer convergence between the two may be possible.

Methods: We undertook a scoping review of literature about the ontological, epistemological and methodological positions of EcoHealth and One Health, and analyzed these according to Lincoln, Lynham and Guba's paradigm framework.

Results: EcoHealth and One Health are both collaborative, systems-focused approaches at the human, animal, and ecosystem health interface. EcoHealth typically leans towards constructivist-leaning assumptions. Many consider this a necessary aspiration for One Health. However, in practice One Health remains dominated by the veterinary and medical disciplines that emphasize positivist-leaning assumptions.

Discussion: The aspirations of EcoHealth and One Health appear to overlap at the conceptual level, and may well warrant closer convergence. However, further shared discussions about their epistemological and ontological assumptions are needed to reconcile important theoretical differences, and to better guide scopes of practice. Critical realism may be a crucial theoretical meeting point. Systems thinking methods (with critical realist underpinnings), such as system dynamics modelling, are potentially useful methodologies for supporting convergent practice.

1. Introduction

EcoHealth and One Health are both holistic, systems-level approaches which sit at the complex interface between human, animal and environmental health interactions. While there are an increasing number of different research approaches located at various levels of this interface, EcoHealth and One Health are the most established named traditions in the field. Both approaches challenge the traditional reductionism of biomedical approaches by locating human and animal health within their wider ecological context (King, 2013; Zinsstag, 2016), and by promoting transdisciplinary and interdisciplinary research (Lapinski et al., 2015; Lebel, 2003; Schelling and Zinsstag, 2015).

Given this shared basis in systems-level thinking, there are increasing calls within the literature for the two approaches to work more closely together in order to address global health challenges. Indeed, a joint One Health and EcoHealth Congress and European One Health/EcoHealth Workshop were each held in 2016 in order to encourage more integrative research between the two (Belgian Community of Practice on Biodiversity and Health, 2016; EcoHealth, 2016). Purported benefits include strengthening both approaches through complementarity; increasing the advocacy power of each; the sharing of resources and costs through economies of scale; and further opportunities for collaboration and knowledge sharing, in order to avoid repetition and gain a more in-depth understanding of the socio-economic and ecological determinants of human, animal and ecosystem health

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(Barrett and Bouley, 2015; Mi et al., 2016; Roger et al., 2016; Zinsstag, 2012). This may be particularly important for dealing with multi-scale, system-wide threats such as pandemics and antimicrobial resistance. More specifically, potential benefits for each approach include applying One Health's use of economic analysis to benefit EcoHealth; further encouraging a focus on animal health in EcoHealth; and more effectively situating One Health research within the context of environmental degradation and over-development (Zinsstag, 2012).

Perceived overlap has led some to conclude that the two approaches will inevitably converge (Charron, 2012b; Mi et al., 2016; Nguyen-Viet et al., 2015), while others looking to a joint "One Health/EcoHealth approach" (Zinsstag, 2016) imply this is already happening at the conceptual level (see also Zinsstag, 2017). However, others argue that important differences between the two remain, requiring further work to reconcile (e.g. Mi et al., 2016).

More specifically, EcoHealth is based on the premise that human health and well-being is dependent on that of the environment (Lebel, 2003). As such, it is primarily concerned with the interactions between humans, ecosystems, and the socio-ecological factors which influence them (Charron, 2012b). EcoHealth is less distinctly focused on the human-animal health interface than One Health (Zinsstag, 2012), instead seeking a broader understanding of health and well-being which spans the humanities and the natural, social and health science disciplines (Leung et al., 2012; Wolf, 2015; Zinsstag et al., 2015).

Evolving from the veterinary and human medicine fields, One Health is primarily concerned with human-animal disease transmission (King, 2013; Lapinski et al., 2015; Zinsstag, 2012). One Health has traditionally been less focused on the role of the environment in contributing to human and animal health (Barrett and Bouley, 2015; Lubroth, 2013). More recently however, its conceptual scope has evolved beyond a narrow focus on disease to include concerns such as food security, climate change adaptation and biodiversity (Evans and Leighton, 2014; Vroegindewey, 2017).

Given their different histories and research traditions, EcoHealth and One Health have developed different practices. Mi et al. (2016) argue that while EcoHealth and One Health are likely to converge over time at the conceptual level, discipline segregation and narrow research approaches currently limit this in practice. Therefore, methodologies may be needed which help to facilitate closer collaboration between the two, where desired. The intention of this paper is not to determine whether closer convergence is needed, but to respond to increasing calls for convergence within the literature. In particular, this study was motivated by the authors' interest in the use of systems thinking and modelling as useful methodologies for supporting both One Health and EcoHealth (as well as convergent) research. However, before an in-depth examination of their potential usefulness can be undertaken, there needs to be a clearer understanding of the underlying assumptions which inform EcoHealth and One Health. This is important given that there remains a lack of clarity regarding the theoretical basis and scope of both approaches due to ongoing conceptual development (Chien, 2013; Evans and Leighton, 2014; Zinsstag, 2016). Singular definitions are elusive, and as such the application of each approach is often varied or context-specific (Lee and Brumme, 2013; Lubroth, 2013; Nguyen et al., 2014).

It is not enough to simply recognize that EcoHealth and One Health have overlapping scopes of practice. Without an understanding of the underlying theoretical positions of EcoHealth and One Health it is unclear both whether meaningful convergence can occur, and whether systems thinking methodologies such as system dynamics modelling are useful for facilitating this. We undertook this review of the theoretical assumptions of EcoHealth and One Health in order to provide a basis from which to inform future convergence and methodological debate. We are not aware of any previous studies that have considered these questions in depth. Therefore, in this study we undertook a systematic scoping review of the literature in order to meet the following objectives:

1. Gain a more in-depth understanding of the ontological, epistemological and methodological underpinnings of both EcoHealth and One Health.
2. Identify areas of difference and overlap between the two approaches.

2. Methods

In undertaking this research, the authors acknowledge their own critical realist perspective. This position, discussed in further detail in the Results section, accepts the existence of an external reality, but that it can only be known subjectively (Bhaskar, 1978).

We conducted our review of the EcoHealth and One Health literature in a systematic way in order to find the full breadth of views that exist. In our review of the literature, our goal was not to seek 'the one truth' about what EcoHealth or One Health 'is'. Instead, we sought to draw together the multitude of views that exist, with no necessary attempt at consensus. Given the calls for convergence in the literature, we wanted to examine the underlying theoretical assumptions of both approaches in order to understand possible points of overlap, but also potential areas of misalignment that indicate convergence may not be desirable, feasible, or requires further philosophical discussion.

2.1. Search strategy

We searched the Scopus database for relevant literature. We chose to focus on Scopus, as it provides comprehensive coverage of research across multiple disciplines, which goes as far back as 1966.

We also included a number of relevant texts that were previously known to the authors and scanned the reference lists of included texts for further relevant literature.

Within Scopus, we performed three searches. The first two searches sought literature that discussed the ontological, epistemological or methodological positions of EcoHealth and One Health respectively, while the third search sought to find relevant literature that compared the two approaches. We used the following search terms:

Search One:

TITLE-ABS ("Eco health" OR ecohealth OR "ecosystem health" OR "ecological health" OR "ecosystem approach* to health") AND (epistemol* OR ontolog* OR "field building" OR methodol* OR philosop* OR framework) AND (LIMIT-TO (LANGUAGE, "English")).

Search Two:

TITLE-ABS ("One health" OR onehealth) AND (epistemol* OR ontolog* OR "field building" OR methodol* OR philosop* OR framework) AND (LIMIT-TO (LANGUAGE, "English")).

Search Three:

TITLE-ABS ("Eco health" OR ecohealth OR "ecosystem health" OR "ecological health" OR "ecosystem approach* to health") AND ("One health" OR onehealth) AND (LIMIT-TO (LANGUAGE, "English")).

There remains some confusion about the use of the term 'ecosystem health' (Connell, 2010). Often, 'ecosystem health' is understood to involve the use of 'health' as a metaphor for assessing the condition of ecosystems (Berbés-Blázquez et al., 2014), without central reference to human health. For the purpose of these searches, the term 'ecosystem health' was used to acknowledge the history of literature that emphasizes the interdependencies of human, animal, and ecosystem health from an anthropocentric position, much of which preceded the emergence of the term EcoHealth in the late 1990s (Webb et al., 2010). We acknowledge that the term 'ecosystem health' as it relates to human health continues to be debated; however, we included the term based on the argument that it is very closely connected to 'ecosystem approaches to health' (de Plaen and Kilelu, 2004), a term itself used synonymously with 'EcoHealth' (Webb et al., 2010).

We also acknowledge that there are likely to be articles which adopted an EcoHealth or One Health approach, but which did not label themselves using the aforementioned terms in the search strings above, and were therefore not included. Our assumption was that those papers which actively reflective on the theoretical assumptions of these approaches would tend to label themselves as such. This is opposed to those studies who may in fact be undertaking EcoHealth or One Health work, but who were not as likely to have engaged in deep reflection on the underlying assumptions of their approach. Additionally, we did not include those studies which labelled themselves as EcoHealth or One Health but did not explicitly discuss their theoretical assumptions, as classifying such studies by their paradigmatic position is often very difficult and vulnerable to misinterpretation.

We included the following source types: journal articles; books; chapters in edited books; commentaries; editorials; and authors' responses published in journals.

The results were limited to those published in English, as translation resources were not available to us for this research. Results were included from 1978 onwards, up until the date the searches were run (1st January 1978–30th June 2017). The date 1978 was chosen as this was when 'ecosystem approaches to health' were first referred to, according to (Webb et al., 2010).

Results from all three searches were included in the review if they:

i. Defined, described and justified the approaches of EcoHealth, ecosystem health, ecosystem approaches to health, or One Health with reference to their underlying theoretical assumptions OR;
ii. Were reviews or meta-analyses of EcoHealth, ecosystem health, ecosystem approaches to health, or One Health OR;
iii. Were technical workgroup reports about the approaches of Ecohealth, ecosystem health, ecosystem approaches to health, or One Health OR;
iv. Were commentaries, editorials, or authors' response letters about the approaches of EcoHealth, ecosystem health, ecosystem approaches to health, or One Health published in journals AND;
v. Were in English

Papers were excluded from the review if they:

i. Were cellular level studies describing basic science research OR;
ii. Were pharmaceutical and clinical trial studies OR;
iii. Were academic curriculum reviews OR;
iv. Were capacity building reports OR;
v. Were studies conducted in languages other than English, as they would have required translation resources which were unavailable OR;
vi. Were articles which used the term 'ecosystem health' to refer to assessment of the condition of ecosystems without making reference to the complex interactions between ecosystem and human health OR;
vii. Were studies relating to Ecohealth, ecosystem health, ecosystem approaches to health, or One Health approaches but which did not discuss the ontology, epistemology or methodology of said approaches

One author (SH) screened the titles and abstracts of all returned results, initially using the exclusion criteria. Potentially relevant papers were reviewed in full by the same author using the inclusion criteria. The same author (SH) classified the included articles by paradigm position, using Lincoln et al.'s (2018) framework (see Section 2.2). A second author (AM) reviewed a sub-set of papers (10%). Consensus was reached between authors regarding the inclusion and exclusion of papers, and the classification of included papers by paradigm position. Outside of the sub-set review, where it was unclear if an article should be included or not, three of the authors (SH, AM and PP) discussed its relevancy and reached a consensus decision. All such discussions, which totaled three, resulted in full agreement. Flow charts detailing the screening process can be found in Appendix A.

2.2. Framework for analysis

In this paper we drew on Lincoln et al.'s (2018) seminal framework of key research paradigms to guide our analysis of the underlying ontological, epistemological and methodological positions of EcoHealth and One Health. We chose this framework, as it is widely adopted and comprehensive in its criteria for categorizing the various paradigm positions according to their ontological, epistemological and methodological assumptions.

Ontology deals with the nature of reality, and our assumptions about the 'real' world (Jonassen, 1991); namely, whether a 'real' world exits independently of our assumptions, or whether reality is socially constructed by individuals as we interact with the world, based on our social and lived experience.

Epistemology deals with the nature of knowledge, including how we acquire knowledge and the claims we make about what constitutes knowledge (Denzin and Lincoln, 2018; Kvanvig, 2005). Ontological and epistemological paradigms exist on a somewhat fluid spectrum with points of overlap and 'interbreeding'. Positivism and constructivism sit at two ends, which are often (but not always) argued to be incommensurable (Lincoln et al., 2018).

According to Lincoln et al. (2018), positivist positions rest upon the ontological assumption that a singular reality exists, external to the knower and independent of humans' subjective experience. In terms of epistemology, knowledge is not considered socially constructed, but is instead a reflection of a singular, external reality (Lincoln et al., 2018). The positivist assumption that a universal, objective 'truth' can be discovered and understood through scientific reasoning forms the basis of much quantitative inquiry (Yilmaz, 2013). Positivism therefore leans heavily towards quantitative methods such as those adopted within epidemiology, which seek to discover 'the' singular, objective 'truth' through the minimization of bias and replicability of results.

In contrast, constructivism's ontological position assumes the existence of multiple realities, which are dependent on the individual and can only be understood through an individual's subjective interpretation (Lincoln et al., 2018). In terms of epistemology, knowledge is considered socially constructed as the knower interacts with the external world and their social, cultural and political environments. Constructivism is particularly suited to qualitative research methods that value the subjective contribution that personal experience, intuition and belief systems make to research (Jonassen, 1991; Max-Neef, 2005; Yilmaz, 2013), for example ethnography and focus group interviews.

In addition to positivism and constructivism, Lincoln et al.'s (2018) framework includes post-positivist, critical theory, and participatory paradigms which sit between positivistism and constructivism on the paradigm spectrum. Positivism and post-positivism are 'positivist-leaning', while critical theory, participatory, and constructivism are 'constructivist-leaning'. Lincoln et al.'s (2018) framework draws these basic paradigm beliefs together, and outlines their position on key research issues (e.g. 'inquiry aim', 'goodness or quality criteria'). Although these more specialized research positions were too specific to be usefully applied to the current EcoHealth and One Health literature, we used them as criteria to help inform our decision about where authors were positioned within the broader categories of ontology, epistemology, and methodology. A summary of this framework, which we used to categorize the included papers, can be found in Table 1.

Table 1 however, misses our own theoretical position (critical realism) and the discussion by Albrecht et al. (2008), which allows for positivism-constructivism to exist as a spectrum with a middle ground

Table 1
Paradigm positions
Adapted from (Lincoln et al., 2018). 'Positivist-leaning' positions sit above the double line; 'constructivist-leaning positions sit below it, and are shaded in grey. Some authors, including Lincoln et al. (2018) argue that positions on either side of the double line are incommensurable (Heron and Reason, 1997).

Paradigm	**Ontology**	**Epistemology**	**Methodology**
Positivism	Assumes the existence of a single reality (that can come to be known through scientific enquiry)	Objective knowledge of the 'real' world can be achieved through scientific enquiry; replicable findings are 'true'	Quantitative, methods, for example many epidemiological studies
Post-Positivism	Assumes the existence of a single reality (that can only come to be known imperfectly)	Objectivity remains an ideal which is sought as closely as possible. Results that are replicated are considered 'probably true'	Primarily quantitative, but can also include qualitative methods
Critical Theory	Reality is shaped by historical, social, political, cultural values over time	Knowledge and researchers' values are informed by historical, social, political, cultural values	Primarily dialectical, qualitative methods which seek to challenge existing value systems and power structures. For example, discourse analysis, critical feminism theory
Participatory (Heron and Reason 1997)	'Reality' is based on experiential interaction with the world; there is a given cosmos with which the mind engages	Knowledge is experiential. It is relative to the knower, and based on their participation with the cosmos	Primarily qualitative methods based on participatory world views, for example participatory action research
Constructivism	Multiple realities exist, and are socially constructed	Knowledge is subjective and 'findings' are constructed through engaging in the research process	Primarily qualitative methods that are dialectical, hermeneutical and seek understanding and reconstruction. For example, discourse analysis, ethnography

position that is ontologically somewhat positivist (there is an external reality 'out there'), but strongly constructivist-leaning in its epistemology and methods (see also Mingers, 2000). Further, as we engaged with the literature, it became clear that there was an important distinction in the way that EcoHealth and One Health were discussed, in terms of the underlying assumptions of each approach in its current form, and the positions that the authors considered each approach could benefit from. Hence, we added 'current form' and 'would benefit from' as sub-categories to the paradigm tables in Appendices B and C.

3. Results

In total, 109 articles were included in the review across all three searches.

Search One produced 1938 results. An additional 4 articles from sources already known to the authors were also accessed. Of these 1942 articles, 1840 were excluded on review of their title and abstract, or due to duplication. Many of these were excluded as they only discussed 'ecosystem health' in relation to the condition of ecosystems. Of the 102 articles reviewed in full, 37 were included.

In Search Two, 1144 results were returned, along with 12 other articles from sources previously known to the authors. Of the total 1156 articles screened, 1019 were excluded on the basis of their title and abstract, or due to duplication. Of the 137 articles reviewed in full, 59 were included.

For the third and final search, 233 results were returned, of which 206 were excluded on review of their title and abstract, or due to duplication. Of the remaining 27 articles, 13 were included upon closer review.

All included articles are listed in Appendix B and C according to where they sat on a paradigm spectrum, based on Lincoln et al.'s (2018) framework. These frameworks demonstrate where the focus within the literature has been. Articles are categorized separately according to their ontological, epistemological, and methodological positions. We also distinguished between discussion of the current paradigm positions of One Health or EcoHealth, and the positions the literature suggested each approach would benefit from adopting.

Articles could appear in more than one category, for example if an article described both the epistemological and methodological positions of One Health, or if they discussed the current versus desired forms of the approach. Articles listed as 'unable to categorize' discussed the underlying paradigms relating to EcoHealth and One Health's ontological, epistemological and methodological positions, but did not clarify these positions enough for us to be able to categorize them. They signal a need for greater consideration and articulation of such aspects by those undertaking reflexive EcoHealth and One Health research.

Overall, methodology was the main focus of the literature included for analysis. There was some discussion regarding the epistemological position of each approach, but very little on their ontological position. Compared to EcoHealth, the One Health literature focused less on its current form than on what is required in order to ensure its effectiveness and future viability, with a large number indicating that One Health needs to more clearly articulate its underlying paradigm and embrace constructivist-leaning methodological values.

3.1. Ontological positions

There was very little discussion regarding the ontological positions of EcoHealth and One Health. Within EcoHealth, one source explicitly discussed the approach in terms of critical theory views of reality (Vansteenkiste, 2014), while Albrecht et al. (2008) defined EcoHealth in terms of critical realism, a post-positivist paradigm that claims ontological realism and epistemological relativism. Several discussed the need for EcoHealth to adopt non-positivist ontological positions (Dakubo, 2010; Kingsley et al., 2015; Vansteenkiste, 2014). Two sources discussed One Heath in terms of positivist-leaning ontological positions (Craddock and Hinchliffe, 2015; Hinchliffe, 2015), and suggested that One Health requires non-positivist views of reality (Craddock and Hinchliffe, 2015; Friese and Nuyts, 2017).

3.2. Epistemological positions

According to the literature, EcoHealth tends to be informed by a constructivist-leaning epistemological position, whereby knowledge is considered socially constructed. EcoHealth approaches typically recognize that research is an exercise in knowledge creation (Charron, 2012a; Dakubo, 2013; Vansteenkiste, 2014), and that researchers themselves are active participants in the generation of subjective ‘truths’ (Charron, 2012b). In this view, knowledge and what is ‘true’ is considered flexible, temporal and contextually-situated.

EcoHealth's lean towards constructivism is informed by its transdisciplinary nature, and in particular its tendency to draw not only on the natural sciences, but also the humanities and social sciences (Charron, 2012c). These fields recognize and actively accommodate human subjectivity and ‘bias’, rather than seeking to minimize it as is common in traditional scientific approaches (Albrecht et al., 2008).

In theory, EcoHealth approaches reject purely positivist claims of a singular, universal ‘truth’ that can come to be known (Charron, 2012b). However, Chimbari (2016) argues that in practice quantitative, typically objectivist approaches continue to be adopted by those undertaking EcoHealth research from a natural sciences perspective. Charron (2012b) argues that EcoHealth draws on the strengths of such approaches where necessary while recognizing more broadly that researchers are involved in the generation of knowledge, meaning purely objective research is not possible.

Some argue that EcoHealth's interest in transformation and empowerment is underpinned by the critical theory assumption that knowledge reflects the power structures within a given society (Dakubo, 2013; Vansteenkiste, 2014). A critical EcoHealth approach therefore situates knowledge within its historical and socio-political context, in order to identify the power dynamics which influence human and ecosystem interactions (Dakubo, 2013). This can involve, for example, adopting feminist theory perspectives to account for human-ecosystem interactions which are determined by socially-dictated gender roles (Lebel, 2003; Vansteenkiste, 2014).

There are those who argue that EcoHealth requires an even greater emphasis on constructivist-leaning conceptions of knowledge, for example understanding social learning (Berbés-Blázquez et al., 2014) and critical theory principles (Dakubo, 2010, 2013).

In contrast to EcoHealth, One Health is dominated by the veterinary and medical sciences (Barrett and Bouley, 2015; Schelling and Hattendorf, 2015). This suggests that the approach continues to adopt those quantitative and epidemiological methods which have traditionally operated within a positivist epistemology, by seeking the minimization of researcher bias and the discovery of universal ‘truths’. It has also been recognized that the epistemological position of One Heath is somewhat dependent on the perceptions held by those engaging in particular research projects (Chien, 2013).

Debate regarding One Health's epistemological scope is largely concerned with whether constructivist-leaning perspectives fit intuitively within the approach. The extent to which One Health accepts that subjective forms of knowledge influence human-animal relationships is central to this discussion. Some argue that One Health is sensitive to the assumption that complex, multiple perspectives of reality exist, and that these conflicting values and truths must be accounted for (Bunch and Waltner-Toews, 2015; Zinsstag et al., 2015). Many others argue that One Health needs to more effectively recognize the heterogeneity of local practices, values and socially constructed ‘realities’ (e.g. Conrad et al. 2013; Hinchliffe, 2015; Wolf, 2015). This, Bardosh (2016) argues, would involve “radically different epistemological perspectives and methods, prompted by the requirement to address complexity, uncertainty and interconnectedness” (p15). Knowledge, in this view of One Health, would recognize that human and animal relationships are in part defined by cultural symbolism (for example, the belief in animals as guardians or spiritual guides) (Ross, 2013), and that divergent ‘truths’ exist regarding the transmission of disease within different cultures or communities (Wolf, 2015). For example, Roth Allen and Lacson (2015) discuss how local understandings of transmission and burial practices hampered efforts to control the West Africa Ebola outbreak.

Similar to EcoHealth, there are those who argue that One Health can and should adopt critical theory insights to examine the power structures and socio-economic conditions which impact on human and animal health within a given context (Craddock and Hinchliffe, 2015; Giles-Vernick et al., 2015). It has also been argued that One Health should embrace the constructivist-leaning understanding that local knowledge can help to facilitate empowered, transformative grassroots solutions (Bardosh et al., 2017).

3.3. Methodological positions

In term of methodological positions, a key aspect of both EcoHealth and One Health is flexibility in the use of methods. As EcoHealth draws upon both the natural and social sciences (Charron, 2012c), it makes use of both qualitative and quantitative methods (e.g. Berbés-Blázquez et al., 2014; Bunch, 2016; Harper et al., 2012; Nielsen et al., 2012; Zinsstag, 2012; Zinsstag et al., 2015). The use of particular methods varies according to the nature of the research being undertaken, although Dakubo (2013) argues that EcoHealth research is nonetheless always built on a constructivist-leaning understanding of values-based knowledge.

In theory, One Health also adopts mixed-method approaches to analyze the human-animal health interface (Schelling and Hattendorf, 2015; Zinsstag and Tanner, 2015). In an isolated example, Woldehanna and Zimicki (2015) used qualitative participatory methods along with a quantitative random sampling survey to better model the factors influencing human-animal disease transmission. Overall however, One Health is dominated by epidemiological studies (Schelling and Hattendorf, 2015) and an interest in quantification (Asokan and Asokan, 2015; Bardosh, 2016; Wallace et al., 2015). Therefore, some argue One Health maintains a ‘black box’ approach towards the social and ecological factors that influence human-animal interactions (Dzingirai et al., 2017a; Lapinski et al., 2015). As such, a number of articles call for greater use of participatory, qualitative, and mixed-methods in One Health to increase its effectiveness (e.g. Cunningham et al., 2017; Dzingirai et al., 2017a; Giles-Vernick et al., 2015; Ossebaard, 2014; Woldehanna and Zimicki, 2015).

According to the literature, another key aspect of both EcoHealth and One Health is that they are characterized by their holistic, systems-level interest in complex, 'wicked' problems at the human, animal and environment interface (e.g. Bunch and Waltner-Toews, 2015; Zinsstag, 2012). More specifically, EcoHealth emphasizes the multi-layered relationships between humans, ecosystems and socio-economic influences (e.g. Lebel, 2003; Saint-Charles et al., 2014; Zinsstag et al., 2015).

One Health too is said to challenge the reductionism of other approaches by accounting for the broader social-economic contexts within which humans exist, occurring across multiple scales (e.g. Conrad et al., 2013; Rüegg et al., 2017). However, some continue to define One Health as a biomedical approach (Capps and Lederman, 2015b), and argue it must do better at addressing the complex, systemic nature of problems in order to live up to its aims and potential (e.g. Bardosh, 2016; Waltner-Toews, 2017). In particular, this includes accounting for the social, political, and economic influences on human-animal interactions (e.g. Craddock and Hinchliffe, 2015; Degeling et al., 2015; Dzingirai et al., 2017b; Kock, 2015).

Both EcoHealth and One Health are characterized by the need for collaboration across different disciplines. Emphasis on transdisciplinary approaches is a key principle of EcoHealth. Transdisciplinary research, in this view, involves a shared problem definition, the integration of methodologies to overcome traditional discipline barriers, and drawing on the knowledge of a range of actors, including policymakers and community (Boischio et al., 2009; Charron, 2012b; Lebel, 2003).

One Health's approach to collaboration appears to be less clearly defined than in EcoHealth. Much of the literature classifies One Health as interdisciplinary (e.g. Degeling et al., 2016; Hinchliffe, 2015; Kingsley and Taylor, 2017; Ross, 2013) but it is also described as being multidisciplinary (e.g. Capps et al., 2015; King, 2013; Lebov et al., 2017), and even transdisciplinary (Rüegg et al., 2017; Ruscio et al., 2015), without a clear underlying paradigm informing either position. Regardless of the label used, there is an identified need to move away from specialized approaches and engage in greater collaboration with the environmental and social sciences, as well as with local communities in order to deliver on One Health's aims of meaningful action (e.g. Degeling et al., 2015; Giles-Vernick et al., 2015; Jenkins et al., 2015).

Beyond mixed-methods, collaboration, and system-level thinking, the other key principles that define EcoHealth include, participation, equity and knowledge-to-action (Charron, 2012c; Lebel, 2003). Embracing these values aligns EcoHealth with constructivist-leaning paradigms that value non-expert knowledge and seek transformation. EcoHealth's interest in equity is tied to social justice values (e.g. Charron, 2012b; Lebel, 2003; Wolf, 2015), and how the health impacts of human-animal-ecosystem interactions fall inequitably on certain members of society, particularly according to gender or social class (Berbés-Blázquez et al., 2014; Dakubo, 2010). EcoHealth's emphasis on active participation is informed by an interest in community empowerment and capacity-building (Charron, 2012a; Dakubo, 2010; Yacoob et al., 2004), allowing locals to contribute to the development of solutions which reflect their priorities as closely as possible (Anticona et al., 2013; Lebel, 2003; Vansteenkiste, 2014). Research of this nature accounts for context-specific knowledge, customs, values, and beliefs systems. For example, SantoDomingo et al.'s (2016) participatory EcoHealth project with two indigenous groups in Colombia drew upon their local ecological, biological and social knowledge to develop EcoHealth calendars that chart specific diseases in relation to annual socioecological changes.

From a constructivist-leaning position, participation in EcoHealth is linked to power-sharing (Boischio et al., 2009), in which the researcher engages in a process of co-learning (Dakubo, 2010). Finally, Charron (2012b) asserts that "change (in various forms) is the intent of ecohealth research" (p12). Knowledge-to-action and the development of practical solutions is widely identified as a key purpose of EcoHealth research (e.g. Bunch, 2016; Lebel, 2003; Patrick and Dietrich, 2016; Zinsstag et al., 2015). However, Charron (2012a) recognizes the challenge of conducting EcoHealth research in practice, in which researchers have to balance a need for empirical approaches and practical, context-specific solutions whose effectiveness is determined by the local community in question.

Key principles of the One Health approach are less clearly outlined than in EcoHealth. Burns and Stephen (2015) argue that One Health is pragmatic by nature, and underpinned by a research-to-action principle which seeks real-world change; Rüegg et al. (2017) further argues that One Health "positions health professionals as agents for change" (p1).

Actors within One Health extend to include non-experts such as local communities and civil society to ensure that practical, effective solutions are developed (Stephen and Stemshorn, 2016; Zinsstag et al., 2015). However, some identify a greater need for community engagement and participatory research approaches in practice (Mackenzie et al., 2014; Min et al., 2013). Adopting a constructivist-leaning position, some argue that a failure to accommodate local communities' concerns and priorities into One Health projects undermines their legitimacy, and disempowers those communities in question (Dzingirai et al., 2017a; Giles-Vernick et al., 2015).

Unlike EcoHealth, One Health does not appear to have equity engrained as a core principle in its approach. There has been some acknowledgement that One Health research can be used to help facilitate equitable outcomes (Cleaveland et al., 2017; Rüegg et al., 2017), however Craddock and Hinchliffe (2015) argue this can only happen once the social sciences are truly integrated into One Health theory and practice.

Despite seemingly deep ontological and epistemological divides, there are considerable overlapping theoretical aspirations across EcoHealth and One Health, particularly in the shared articulation of the need for systems thinking, and in the recognition among One Health theorists for more critical, constructivist approaches to enrich understanding and enable change to occur. From the above analysis, we argue that finding the fertile ground for rich convergent research will require methodologies that are compatible with critical realism and can address the need for participatory systems thinking, while allowing for combinations of qualitative and quantitative analysis. In the following section we therefore discuss system dynamics modelling [SDM] as a potential convergence methodology that could meet these criteria.

4. Discussion

In this paper we conducted a systematic scoping review of EcoHealth and One Health literature in order to identify the underlying theoretical assumptions and area of overlap between the two approaches. This was done with the aim of better understanding whether closer convergence between the two may be possible. Further, this review was undertaken with the intention of providing an in-depth basis from which to inform future research and debate regarding the usefulness of systems thinking methodologies for facilitating this convergence.

As system-level approaches, both EcoHealth and One Health are based on the theoretical assumption that interactions between humans, animals and their wider ecosystems involve complex, non-linear relationships and feedback loops at various scales. There have been calls for closer convergence of the two in the literature, as well as for both to aspire to more participatory and collaborative approaches and mixed-methods in practice. To achieve this, systems-focused methodologies that transcend traditional discipline silos and can accommodate qualitative and quantitative methods may be needed. This is particularly important given the diverse range of methods and worldviews adopted by those who conduct research under the respective banners of EcoHealth and One Health. It is in this context that it is useful to consider whether systems thinking methodologies such as SDM may be useful to support such efforts. SDM aims to understand non-linear relationships within complex systems by using arrows to visually map feedback loops and changing relationships over time, often using computer simulation (Forrester, 1993).

It is important to note that we are limited in the inferences we can make about the full spectrum of theoretical assumptions within both approaches, given that this review was limited to articles which purposively reflected on these assumptions, rather than including all papers using EcoHealth or One Health terms in practice. However, our results suggest that EcoHealth and One Health have emerged from very different research traditions and continue to lean, respectively, towards constructivist-leaning and positivist-leaning values. According to the literature, EcoHealth tends to draw on constructivist-leaning principles that seek empowerment and transformation through active participation, an emphasis on equity, and methods that value the subjective nature of knowledge. One Health appears to be less clearly defined at the theoretical level. Half of the included articles discussing One Health we deemed 'unable to categorize' by paradigm position. In practice, however, One Health research remains relatively limited to the animal and medical sciences, with leanings towards positivist paradigms. There are strong calls for One Health to better embrace the social sciences and to more strongly value human subjectivity and active participation in order to fulfil its aims.

Based on the assumption of Lincoln et al.'s (2018) framework that positivist and constructivist-leaning paradigms are incommensurable, this would suggest that the two approaches are in fact incompatible. However, it is at this point that we may begin to consider whether critical realism offers a potential theoretical meeting point, and further, whether systems thinking methodologies such as SDM offer a potentially useful convergence methodology. Some authors, such as Albrecht et al. (2008), already claim that EcoHealth adopts a critical realism position, and this is reminiscent of other approaches which challenge the strict incommensurability of traditional paradigmatic frameworks (e.g. pragmatism Tashakkori and Teddlie, 1998). Likewise, Pruyt (2006) argues that SDM as a methodology cannot be defined by a strict positivist/constructivist divide, but rather is informed by a critical realism paradigm position. SDM therefore accepts that an external reality exists, while accommodating the focus on subjectivity and active participation (Bala et al., 2017; Stave, 2002) that is required in EcoHealth and called for in One Health. Further, SDM offers the potential to accommodate key needs in both approaches by focusing on systems-behavior, using both qualitative and quantitative methods, and accounting for the political, economic, and social influences within systems (Bala et al., 2017). In particular, participatory SDM may facilitate equitable collaboration by sharing ownership of a research project with local community and other key stakeholders. SDM therefore appears to offer a potentially useful way to accommodate the principles of collaboration, participation and mixed-methods that many argue EcoHealth and One Health both aspire to. Based on our examination of the underlying assumptions of EcoHealth and One Health, we encourage further debate about the philosophical underpinnings of closer convergence between the two, as well as an in-depth exploration of the potential for systems thinking and modelling, in particular SDM, as possible methodologies for facilitating it.

In this paper, we included articles which self-identified with 'One Health', 'EcoHealth', 'ecosystem approaches to health', or 'ecosystem health', and did not assess whether they aligned with widely accepted definitions of these approaches. This may have led to the inclusion of articles that may not otherwise be considered either EcoHealth or One Health research, as well as the exclusion of (possibly) relevant articles that did not label their approaches as such.

We also acknowledge that there are an increasing number of approaches focussed at various levels of human-animal-environment health interactions which we have not included in our analysis, such as the approach recently articulated as Planetary Health and older approaches, such as Conservation medicine. We chose to focus on One Health and EcoHealth as they are the two major, long-standing named traditions within the field, and because there have been increasing calls in the literature and in practice for closer convergence between these two approaches specifically.

Since we ran our searches, a few relevant articles have been published which summarize the EcoHealth and One Health approaches (e.g. Buse et al., 2018; Lerner and Berg, 2017; Oestreicher et al., 2018). These summaries support the key principles of each approach that we have identified here. Additionally, they indicate the need to consider in more detail the core philosophical assumptions that underpin both approaches, as well as areas of difference, overlap, and possible convergence. As far as we are aware, no other studies have systematically reviewed and compared the theoretical assumptions of EcoHealth and One Health, nor considered the use of SDM as a possible convergence methodology.

Despite the comprehensive nature of Lincoln et al.'s (2018) framework, we deemed a number of articles 'unable to categorize'. This included, for example, articles which discussed the need for approaches which were both typically positivist-leaning (e.g. greater disease surveillance), and constructivist-leaning (e.g. community participation and the inclusion of social values) without clarifying whether they sat within a positivist paradigm, or a constructivist-leaning one (e.g. Mackenzie et al., 2014). There were also more clearly defined positions that nonetheless did not align with Lincoln et al.'s (2018) framework. For example, Albrecht et al.'s (2008) critical realism approach explicitly discussed EcoHealth in terms of a positivist-leaning ontological position, and constructivist-leaning epistemology positions on human subjectivity and culturally-situated knowledge. We distinguished between Albrecht et al.'s (2008) ontological and epistemological position because of this explicit distinction that they made, despite the fact that these paradigms are often considered incommensurable. Despite Lincoln et al.'s (2018) adherence to the incommensurability of positivist and constructivist-leaning paradigms, their work remains a useful foundation for the basis of our scoping review, given that the EcoHealth and One Health literature was generally unclear about their underlying assumptions. We therefore deemed it more practical to use Lincoln et al.'s (2018) framework for initially guiding our categorization of the two approaches according to their ontological, epistemological and methodological positions.

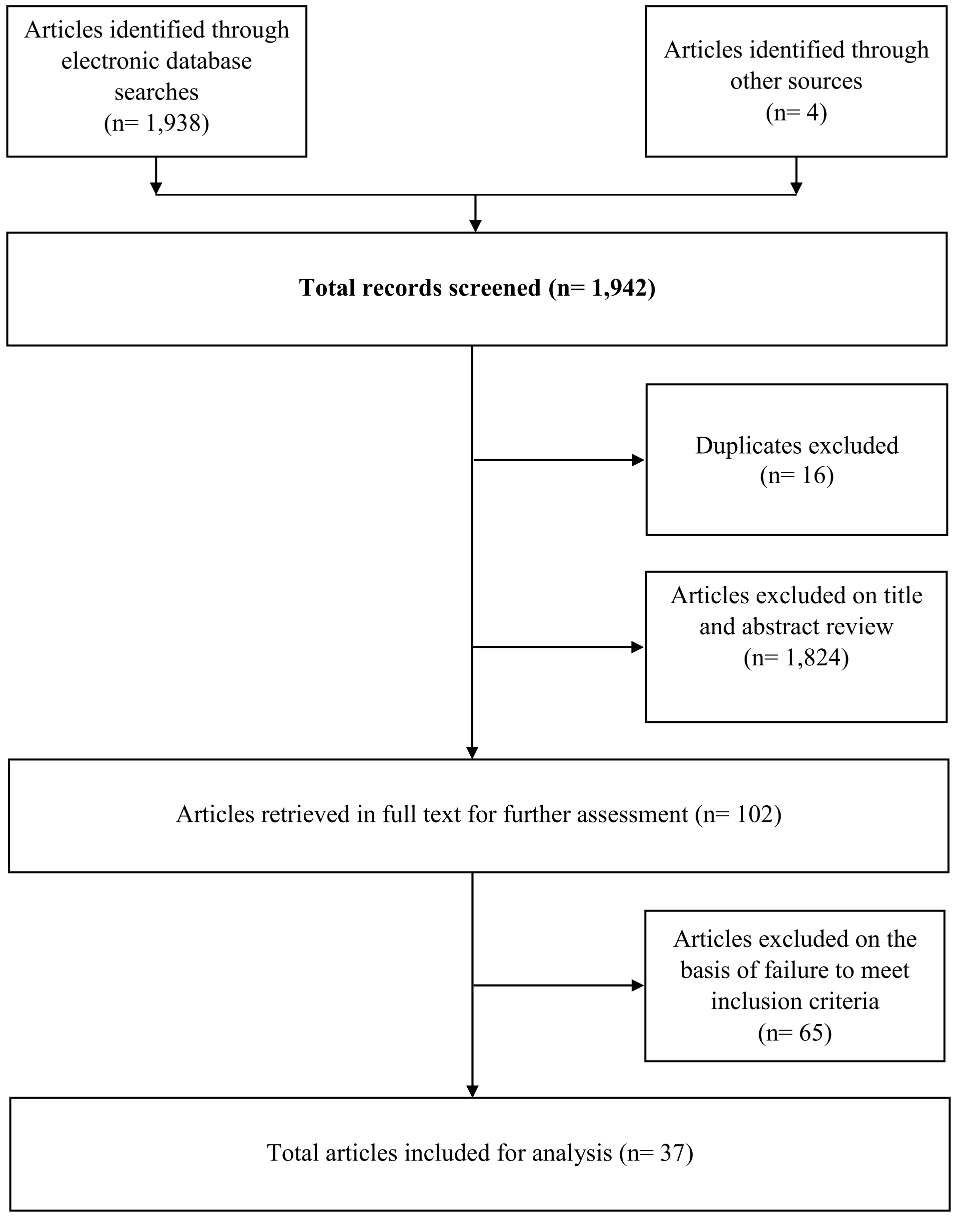
There appears to be considerable overlap in the theoretical aspirations of EcoHealth and One Health, particularly in the need for integrative, systems thinking research, and in calls for more critical, constructivist approaches within One Health research. Closer convergence between the two offers potential benefits such as the sharing of resources, knowledge, costs and advocacy efforts. However, EcoHealth and One Health researchers need to more clearly articulate their theoretical assumptions and actively discuss where theoretical common ground might lie. Clearer understanding of the aims and scope of both approaches would shed further light on both the desirability and feasibility of closer convergence, particularly given the diverse nature of research and methods being used by EcoHealth and One Health researchers in practice. Further, it would help to inform decisions about the use of SDM as a methodology to help facilitate this convergence. As it stands, we suggest that there is potential to more closely integrate EcoHealth and One Health research which does not adhere to a strict positivist-constructivist divide, but rather incorporates or draws on a diverse range of methods and worldviews. In these instances, systems thinking approaches such as SDM may be potentially useful methodologies for integrating these approaches in a way that is beneficial for both.

Declaration of Competing Interests

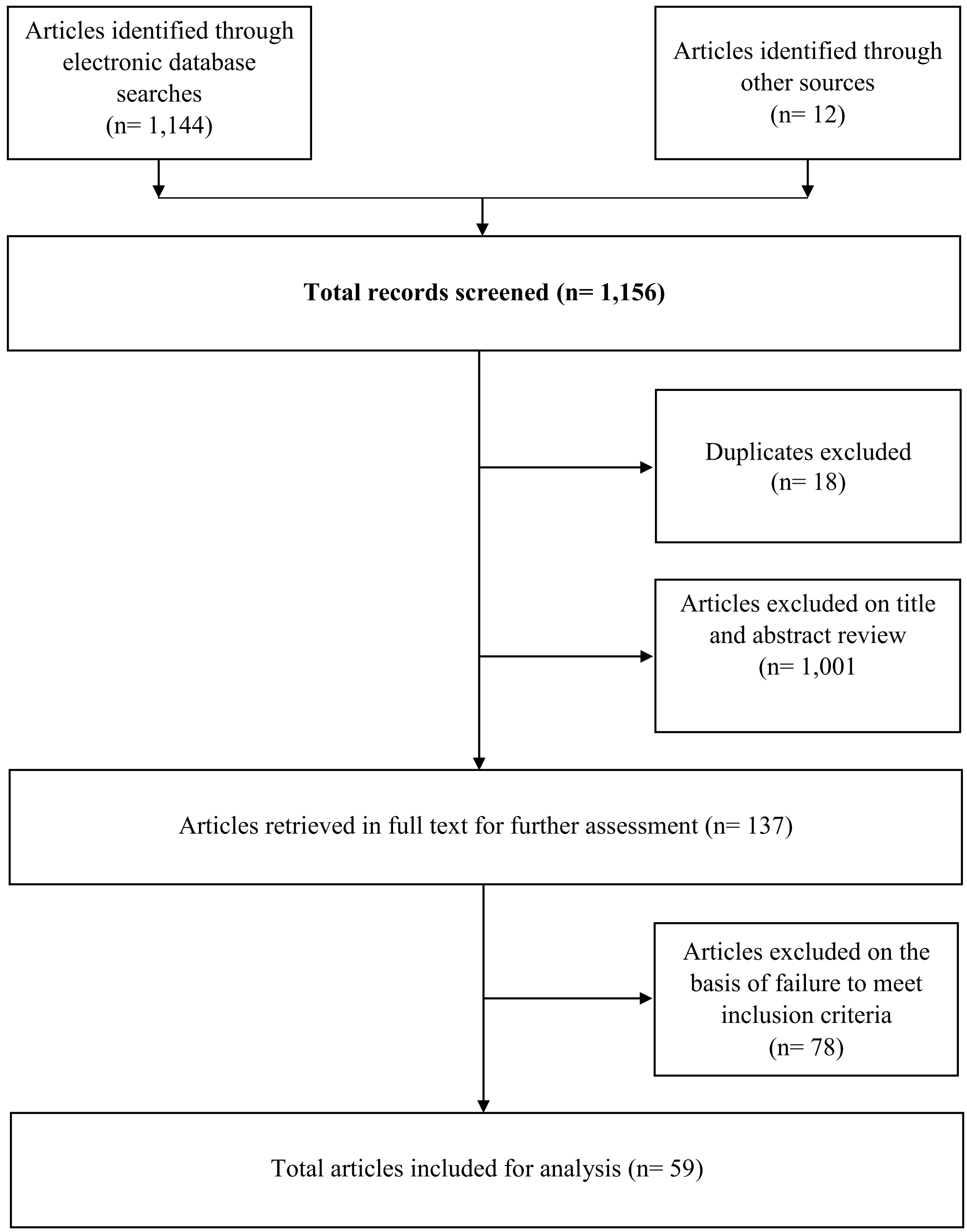
The authors declare they have no actual or potential competing financial interests.

Appendix A. Search flowcharts

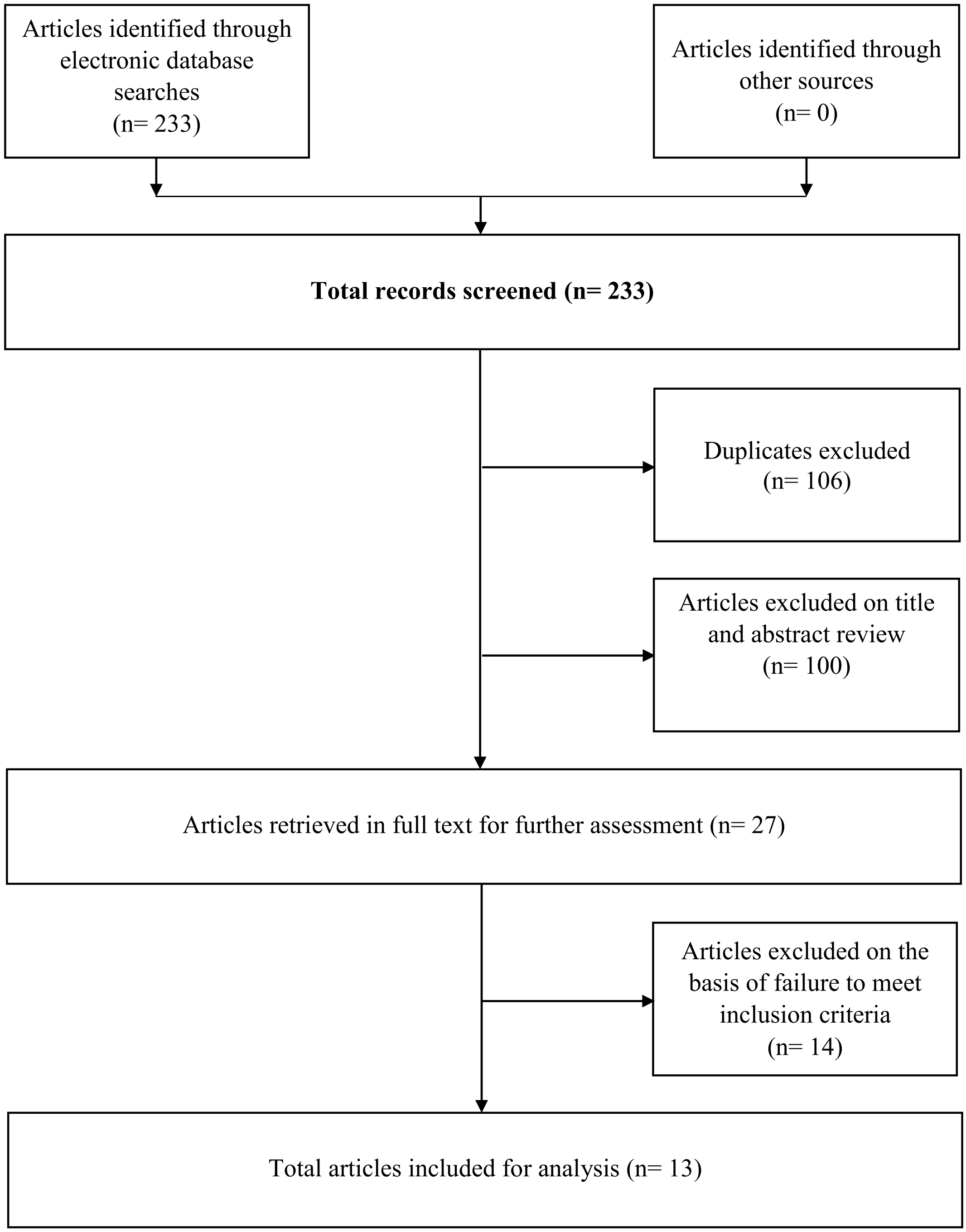
Search One flow chart (EcoHealth)



Search Two flow chart (One Health)



Search Three flow chart (EcoHealth and One Health)



Appendix B. One Health: included articles by paradigm position

Articles are listed according to a spectrum based on Lincoln et al.'s (2018) paradigm framework. Distinctions are made between ontological, epistemological and methodological positions, as well as what the literature considers to be One Health's current form, and the proposed positions recommended to improve current practice. Articles assigned to the 'unable to categorize' columns did not discuss the underlying paradigms in relation to One Health in enough detail for us to be able to categorize.

Paradigm spectrum	Ontology		Epistemology		Methodology	
	One Health (current form)	One Health (would benefit from)	One Health (current form)	One Health (would benefit from)	One Health (current form)	One Health (would benefit from)
Constructivist-leaning (Constructivism, Participatory, Critical)		(Craddock and Hinchliffe, 2015; Friese and Nuyts, 2017; MacGregor and Waldman, 2017)	(Ruscio et al., 2015)	(Bardosh, 2016; Bardosh et al., 2017; Conrad et al., 2013; Craddock and Hinchliffe, 2015; Degeling et al., 2015; Giles-Vernick et al., 2015; MacGregor and Waldman, 2017; Mwacalimba and Green, 2015; Ross, 2013; Wolf, 2015)	(Rüegg et al., 2017)	(Bardosh, 2016; Bardosh et al., 2017; Capps and Lederman, 2015b; Charron, 2012b; Conrad et al., 2013; Craddock, 2015; Craddock and Hinchliffe, 2015; Cunningham et al., 2017; Degeling et al., 2015; Dzingirai et al., 2017a; Dzingirai et al., 2017b; Friese and Nuyts, 2017; Giles-Vernick et al., 2015; Kock, 2015; MacGregor and Waldman, 2017; Min et al., 2013; Quinlan and Quinlan, 2016; Ross, 2013; Rüegg et al., 2017; Ruscio et al., 2015; Stephen and Stemshorn, 2016; Walther et al., 2016; Waltner-Toews, 2017; Whittaker, 2015; Wolf, 2015)
Unable to categorize				(Bunch and Waltner-Toews, 2015; Zinsstag et al., 2015)	(Bardosh et al., 2017; Baum et al., 2017; Binot et al., 2015; Burns and Stephen, 2015; Capps et al., 2015; Capps and Lederman, 2015a, 2015b; Chien, 2013; Cleaveland et al., 2017; Cunningham et al., 2017; Degeling et al., 2016; Hinchliffe, 2015;King, 2013; Lapinski et al., 2015; Leung et al., 2012; Lysaght et al., 2017; Mackenzie et al., 2014; Mardones et al., 2017; Mi et al., 2016; Ossebaard, 2013; Ross, 2013; Ruscio et al., 2015; Schelling and Hattendorf, 2015; Schelling and Zinsstag, 2015; Stephen and Stemshorn, 2016; Woldehanna and Zimicki, 2015; Zinsstag, 2012, 2016; Zinsstag and Tanner, 2015; Zinsstag et al., 2015)	(Aguirre et al., 2016; Binot et al., 2015; Capps and Lederman, 2015a, 2015b; Cleaveland et al., 2014; Cumming and Cumming, 2015; Degeling et al., 2016; Hall and Durrheim, 2011; Hattendorf et al., 2017; Hueston et al., 2013; Jenkins et al., 2015; Kingsley and Taylor, 2017; Lapinski et al., 2015; Leung et al., 2012; Mackenzie et al., 2014; Mardones et al., 2017; Mi et al., 2016; Narrod et al., 2012; Ossebaard, 2013; Ossebaard, 2014; Schelling and Hattendorf, 2015; Schelling and Zinsstag, 2015; Stenberg and Chisholm, 2012; Wallace et al., 2015; Woldehanna and Zimicki, 2015; Zinsstag, 2016; Zinsstag and Tanner, 2015; Zinsstag et al., 2015)
Positivist leaning (Post-positivist, Positivist)	(Craddock and Hinchliffe, 2015; Hinchliffe, 2015)		(Asokan and Asokan, 2015)		(Aguirre et al., 2016; Asokan and Asokan, 2015, 2016; Bardosh, 2016; Barrett and Bouley, 2015; Heymann and Dixon, 2013; Kingsley and Taylor, 2017; Kock, 2015; Lebov et al., 2017; Loh et al., 2013; Lubroth, 2013; Manlove et al., 2016; Narrod et al., 2012; Wallace et al., 2015)	(Barrett and Bouley, 2015; Heymann and Dixon, 2013; Rabinowitz et al., 2017)

Appendix C. EcoHealth: included articles by paradigm position

Articles are listed according to a spectrum based on Lincoln et al.'s (2018) paradigm framework. Distinctions are made between ontological, epistemological and methodological positions, as well as what the literature considers to be EcoHealth's current form, and the proposed positions recommended to improve current practice. Articles assigned to the 'unable to categorize' columns did not discuss the underlying paradigms in relation to EcoHealth in enough detail for us to be able to categorize.

Paradigm spectrum	Ontology		Epistemology		Methodology	
	EcoHealth (current form)	EcoHealth (would benefit from)	EcoHealth (current form)	EcoHealth (would benefit from)	EcoHealth (current form)	EcoHealth (would benefit from)
Constructivist leaning (Constructivism, Participatory, Critical)	(Vansteenkiste, 2014)	(Dakubo, 2010; Kingsley et al., 2015; Vansteenkiste, 2014)	(Albrecht et al., 2008; Anticona et al., 2013; Charron, 2012a, 2012b; Dakubo, 2013; Min et al., 2013; Richter et al., 2015; Vansteenkiste, 2014)	(Berbés-Blázquez et al., 2014; Dakubo, 2010, 2013)	(Albrecht et al., 2008; Anticona et al., 2013; Berbés-Blázquez et al., 2014; Boischio et al., 2009; Bunch et al., 2011; Bunch, 2016; Charron, 2012a, 2012b, 2012c; Chimbari, 2016; Craddock and Hinchliffe, 2015; Dakubo, 2010, 2013; de Jesus Lawinsky et al., 2012; Grace et al., 2012; Harper et al., 2012; Harper et al., 2015; Jenkins et al., 2015; Kingsley et al., 2015; Lebel, 2003; Leung et al., 2012; Mi et al., 2016; Min et al., 2013; Neudoerffer et al., 2005; Patrick and Dietrich, 2016; Richter et al., 2015; SantoDomingo et al., 2016; Vansteenkiste, 2014; Waltner-Toews, 2009; Webb et al., 2010; Wolf, 2015; Yacoob et al., 2004; Zinsstag, 2012)	(Anticona et al. 2013; Berbés-Blázquez et al., 2014; Chimbari, 2016; Dakubo, 2010, 2013; Grace et al., 2011; Kingsley et al., 2015; Neudoerffer et al., 2005; Nguyen et al., 2014; Richter et al., 2015; Wolbring, 2014)
Unable to categorize			(Stephen et al., 2016)		(Bevilacqua et al., 2015; Bunch and Waltner-Toews, 2015; Burns and Stephen, 2015; Crawshaw et al., 2014; Grace et al., 2012; Neudoerffer et al., 2005; Nguyen-Viet et al., 2015; Nielsen et al., 2012; Parkes, 2011; Possas, 2001; Saint-Charles et al., 2014; Stephen et al., 2016; Zinsstag, 2016; Zinsstag et al., 2015)	(Bevilacqua et al., 2015; Nielsen et al., 2012; Parkes, 2011; Possas, 2001; Stephen et al., 2016; Zinsstag, 2016)
Positivist leaning (Post-positivist, Positivist)	(Albrecht et al., 2008)				(Burger and Gochfeld, 2001; Chimbari, 2016)	

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